KARAN BHORIYA

AI Engineer

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LinkedIn | GitHub | Portfolio

Professional Summary

AI Engineer driven by optimizing end-to-end intelligent systems. Focused on building production-grade autonomous workflows that fuse NLP, Large Language Models (LLM), Retrieval-Augmented Generation (RAG) systems, and agentic AI innovation with computational efficiency to solve high-impact problems at scale.

EDUCATION

Sushant University, Gurugram **2023 – Present**
B.Tech, Computer Science Engineering (Specialization in AI & ML)

INTERNSHIP

AI/ML Engineer -Sopra Steria, Noida **May 2026 – Aug 2026**

- Built Enterprise SAP OData Gateway (FastAPI, LangGraph) with Hybrid Knowledge Graph-RAG via SurrealDB, reducing data retrieval time from hours to sub-5 seconds.
- Developed Ephemeral Air-Gapped Python Sandbox (Docker, AST linting) for automated SAP data extraction, cutting enterprise reporting latency by 90%.
- Deployed Next.js 15 conversational interfaces with real-time SSE streaming (Web/Telegram/Teams), reducing LLM token consumption by 85%.

Prompt Engineer & Data Analyst -Humana People to People India, Vasant Kunj **Apr 2025 – Aug 2025**

- Engineered Python validation pipelines and automated schema checks, reducing QA time from 4 days to 2 hours.
- Applied LLM prompt engineering to automate data cleaning and geographic code-matching utility 87% faster than human.
- Developed Power BI dashboards translating complex data pipelines into actionable insights for stakeholders.

PROJECTS

Scholar Forge -Multi-Agent Academic Paper Pipeline **AUG 2026**

- Built 9-agent system (LangGraph, FastAPI, PostgreSQL + pgvector) that converts topic + uploads into submission-ready paper with live arXiv, Semantic Scholar, OpenAlex verification..
- Engineered ingestion, citation verifier (DOI resolution, fuzzy match), and LaTeX normalizer + ReAct repair loop.
- Deployed Dockerized stack with SSE streaming UI, human checkpoints, PDF/Word/LaTeX outputs in 4-8 min per paper with live demo on Vercel.

TechHub AI -Multi-Agent Swarm Debate Platform **Jun 2026**

- Built multi-agent debate system (FastAPI, Next.js, AgentScope): 4 different persona agents debating tech trade-offs and a judge synthesizing verdict.
- Achieved 95% token reduction via research pre-summarization and sliding-window compression 3K-5K vs 100K+ tokens.
- Deployed synthesis engine scoring 3 tech stacks; 10x cost & latency reduction; live on Vercel.

Career-Orbit -Autonomous Multi-Agent Job Hunter **May 2026**

- Built 6 AI agents (AgentScope, Python, Playwright) crawling LinkedIn/Wellfound etc, scoring CV-fit 0-100, drafting recruiter emails.
- Integrated SurrealDB Graph RAG for local-first tracking zero cloud, zero crawling cost; human-in-the-loop form filling via Playwright.
- Deployed via Docker with live demo on Vercel.

TalentAI -AI-Powered Recruitment & HR Ecosystem **Feb 2026**

- Built AI recruitment platform (Python, Flask) with Gemini Pro & Groq Llama 3.1 for PDF resume parsing and scoring.
- Weighted ranking (skills 50%, exp 30%, academics 20%) + full CRUD HR workflows and applicant tracking.
- Live hiring dashboards with ranking for data-driven screening.

Additional 10+ projects on Github

PUBLICATIONS

When Should an AI Agent Say "I Don't Know"? Evaluating Abstention and Reliability in a Financial Support RAG Agent **Under Review**

- Built a behaviour-first benchmark of 150 banking queries across 5 types, scoring each reply on 5 expected acts (answer, clarify, abstain, escalate, refuse) in a fixed RAG pipeline.
- Compared 4 LLM configurations (DeepSeek and GLM, each at high and low reasoning effort) over 600 replies using 3-checker majority voting (Fleiss kappa 0.62); the overall miss rate was 19.5% with no significant pairwise gap after Holm correction.
- Found misses concentrate on ambiguous queries (11-15 of 30) rather than adversarial ones (1-3 of 30), with "failure to answer" the top reason (51 of 117); open-sourced the queries, codebook, and analysis scripts on GitHub.

High-Fidelity Digital Twins: Explainable AI for Fleet-Wide Battery State-of-Health Prognostics in Industry 4.0 **Accepted**

- Pioneered explainability-first digital twin combining 2-layer Attention-LSTM with TimeSHAP pruning (83% reduction) for SoH forecasting and sensor attribution.
- Achieved RMSE 0.0359 and R2 0.64 on NASA Dataset; engineered 5-feature pipeline with train-only scaling and no future leakage.
- Designed Streamlit and Docker deployment with single-value prediction and dual explanations (attention + SHAP) for critical energy infrastructure.

Edge-Native Fault Detection in Smart Inverters via End-to-End Graph Optimization **Accepted**

- Pioneered a novel, first-of-its-kind architecture combining physics-informed deep autoencoders, monolithic ONNX graph deployment, and INT8 quantization for real-time edge fault detection on solar smart inverters.
- Telemetry preprocessing pipeline (5-step rolling windows, custom feature extraction, weighted MSE + safety boundaries).
- Achieved 97.75% accuracy & 92.45% F1-Score; 72.34% model compression; 0.0686 ms latency; 16.98% CO2 reduction.

Human-Centered Emotion Classification Using Classical Machine Learning Models on the GoEmotions Dataset **Under Publication**

- Comparative study: classical ML models vs Transformers (RoBERTa, DistilBERT) on 58K Reddit comments.
- NLP preprocessing pipeline (redundancy reduction, Ekman mapping, spaCy, TF-IDF/CountVectorizer).
- Classical models achieved 53% accuracy & 43% Macro F1-Score, rivaling Transformers while reducing latency by 120x.

EXTRACURRICULAR ACTIVITIES

Community Service Director -Rotaract Club of Sushant University **Aug 2024 – Jul 2025**

Member -Rotaract Club of Sushant University **Aug 2023 – Jul 2024**

SKILLS

Languages & Frameworks: Python, Docker, AgentScope, CrewAI, LangGraph, Advanced RAG, PostgreSQL, Crawl4AI, PyTorch, TensorFlow, Keras, Scikit-Learn, spaCy, Pandas, NumPy, Matplotlib, Streamlit, FastAPI, Flask, ONNX, Pydantic, SurrealDB, PostgreSQL, Git, Power BI, Jupyter.